MSIS-2419 — Business Intelligence and Data Warehousing

Santa Clara University · Leavey School of Business · Department of Information Systems & Analytics — Graduate Program
Winter Quarter 2027 · 4 units · 10 weeks, 20 sessions

This syllabus collects, in one document, what the course repository publishes in its separate pages. If this page and a linked page ever disagree, the linked page in `course_information/` is the one that applies, and the instructor will correct this page.

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1. Course Information

Course	MSIS-2419 — Business Intelligence and Data Warehousing
Units	4 (graduate)
Quarter	Winter 2027: January 4 – March 20, 2027
Format	10 weeks · 20 sessions (2 per week) · 95 minutes per session, each with a 30-minute in-class lab
Meetings	Mondays and Wednesdays, 5:45–7:20 pm (Pacific Time) · first class Mon January 4, last class Wed March 17, 2027 · no class on Mon Jan 18 (Martin Luther King Jr. Day) and Mon Feb 15 (Presidents' Day)
Classroom	*To be announced on Camino*
Prerequisites	ISBA 2401 (Data Analytics with Python) and ISBA 2402 / MSIS-2402 (Database Management Systems — Fundamentals of SQL) — see Prerequisites
Learning management system	Camino (Canvas) — announcements, Discussions, gradebook
Course materials	Course repository: https://github.com/mahmoudparsian/data-warehousing — or one ZIP per week (`DOWNLOADING_COURSE_FILES.md`)

Instructor

Instructor	Dr. Mahmoud Parsian (`instructor.md`)
Email	mparsian@scu.edu
Office	Lucas Hall, Room 316U
Office hours	*To be announced on Camino* (Zoom, and by appointment) — see `office_hours.md`
Teaching assistant	*To be announced*

All quarter dates are kept in one page: Class Meeting Dates & Hours.

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2. Course Description

From the SCU catalog:

> Introduces the business, technology, and managerial issues related to BI and DW solutions. Students will acquire practical skills in collecting business requirements, planning, defining, designing and developing a BI solution. Emphasis is placed on learning how to derive business value from BI and DW solutions. Hands-on experience will be obtained using a variety of BI tools.

A data warehouse is a **subject-oriented, integrated, non-volatile, and time-variant** collection of data in support of management's decisions (Bill Inmon). This term the course moves through four parts:

1. **SQL for analytics** (weeks 1–2): business questions → SQL → insight; `GROUP BY`, `HAVING`, `CASE`, window functions, and exploratory data analysis (EDA) with SQL.
2. **Classic data warehousing** (weeks 3–4): OLTP vs. OLAP, facts, dimensions and grain, ETL, star schemas, slowly changing dimensions, OLAP operations (`ROLLUP`, `CUBE`, `GROUPING SETS`), and a first Tableau dashboard.
3. **Medallion architecture** (weeks 5–8): Bronze (land everything, with lineage), Silver (clean, conform, integrate several sources), and Gold (a star schema, KPIs, data marts), on files (a lakehouse) and in a database.
4. **Wrap-up** (weeks 9–10): incremental pipelines (load logs, late data, corrections, proving incremental = full rebuild), KPIs that add up, dashboards, data storytelling.

One running case study: a pizza shop's sales data, from raw files (weeks 6–7) through a Gold star schema (week 8) and monthly incremental loads (week 9) to the final data story (week 10). Every session has a hands-on lab in **DuckDB** with **Marimo** notebooks; dashboards use **Tableau** or Python charts. Details: `course_info_and_desc.md`.

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3. Learning Outcomes

By the end of the course, students can:

1. Translate business questions into analytical SQL (aggregation, `CASE`, window functions) and explain the results
2. Profile a dataset (EDA) and turn its problems into explicit data-quality rules
3. Design a star schema: grain, facts, dimensions, surrogate keys, slowly changing dimensions
4. Build ETL / ELT pipelines that are traceable (lineage) and reconcile every row
5. Implement the Medallion architecture (Bronze, Silver, Gold) on files and in a database
6. Conform and integrate data from several sources (mapping tables, time zones, deduplication)
7. Load data incrementally and idempotently, handle late data and corrections, and prove the result
8. Write OLAP queries (roll-up, drill-down, slice, dice, pivot, `ROLLUP` / `CUBE` / `GROUPING SETS`)
9. Define KPIs correctly (additive vs. non-additive measures) and build data marts for business users
10. Build dashboards (Tableau, Python) and present a data story with a recommendation

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4. Course Schedule

Two sessions per week; every session has a **30-minute in-class lab** (20 labs). Week pages, slides, notebooks and labs: Weekly schedule — start with Week 0: course outline and software setup.

Week	Topic	Labs
1	Foundations of SQL for Analytics	1.1 From Business Questions to SQL · 1.2 `GROUP BY`, `HAVING` and `CASE`
2	SQL Analytical Thinking & Exploratory Data Analysis	2.1 Window Functions · 2.2 EDA with SQL
3	Classic Data Warehousing & ETL	3.1 Your First Star Schema · 3.2 ETL: Extract, Clean and Load
4	Star Schema in Depth & OLAP	4.1 Slowly Changing Dimensions · 4.2 OLAP on a Star Schema (and Tableau)
5	Medallion Architecture: Introduction	5.1 Your First Medallion Pipeline · 5.2 Medallion as a Lakehouse
6	Bronze in Depth & Silver Cleaning	6.1 Bronze in Depth · 6.2 Silver Cleaning
7	Silver: Conforming & Integrating Sources	7.1 Conforming a Second Source · 7.2 Integrating Two Sources
8	Gold Layer & Star Schema	8.1 Gold: A Star Schema · 8.2 Gold KPIs and Data Marts (and Tableau)
9	End-to-End Pipeline, Incremental Processing & Project Workshop	9.1 An Incremental Pipeline · 9.2 Late Data, Corrections, Proving the Pipeline Right
10	Final Project & Data Storytelling	10.1 From Gold to a Data Story · 10.2 Dashboard and One-Slide Story
Bonus	*Optional:* LLM + SQL Integration	Bonus module

Assessment	When
Midterm Exam	*To be announced on Camino* — during a class session (`midterm_exam.md`)
Exploratory data presentation (project)	*To be announced on Camino* (Project guidelines)
Final project presentations	Week 10 (Project guidelines)
Final Exam	*Finals week, to be announced on Camino* (`final_exam.md`)

5. Required Materials

Books (`required_books.md`):

1. *Data Warehousing Tutorial* — TutorialsPoint (online version)
2. *Building the Data Lakehouse* — Bill Inmon, Mary Levins, Ranjeet Srivastava (PDF)

Software — set up before the first lab, following Week 0: software setup (it includes a notebook that checks your installation). Full list: `required_software.md`.

Tool	Used for
DuckDB **2.0 or newer** — the CLI and the Python package (`pip install "duckdb>=2.0"`)	The database for the whole course; keep one DuckDB version for both
Python **3.11+** with pandas, matplotlib, seaborn (week 2 EDA: openpyxl, scipy, plotly)	Notebooks and charts

Tool	Used for
Marimo, with `sqlglot` (`pip install marimo sqlglot`; sqlglot runs the SQL cells)	All course notebooks and the in-class lab notebooks
Tableau Desktop (free with a Tableau for Students license) or Tableau Public (free)	Dashboards from week 4 (the same charts in Python are accepted)
Respondus LockDown Browser (SCU download)	Midterm and Final Exam
Optional: MySQL; an OpenAI API key	MySQL reference examples; the optional bonus module

A laptop with the software above, at every session — every session has an in-class lab.

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6. Assessment and Grading

Component	% of grade
In-class labs — 20 labs (2 per week); the lowest 2 are dropped	40
Project	30
— Project documentation / exploratory data analysis	*10*
— Project implementation	*10*
— Project presentation	*10*
Midterm Exam	10
Final Exam	20
Total	**100**
Bonus — for exceptional work, extra credit added on top of the 100	up to +2

Labs

- 30 minutes each, done **in class**: a lab sheet and a Marimo notebook, released at the start of the session. Submit your work as instructed in class.
- The lowest 2 of the 20 lab scores are dropped.

Project

In teams of 2, students design, build and present a data warehouse on a dataset of their choice. Final presentations are in week 10, in both sessions: 10 minutes + 3 minutes of questions per team (each team is assigned to a session). Details: Project guidelines.

Exams

The midterm and final exams are **closed book, closed notes, closed software, closed Internet, closed phone**, and taken in the **Respondus LockDown Browser**. No makeup exams are given.

Letter grades

Your course percentage (including any bonus, so at most 102%) is rounded to the nearest tenth before this table is applied: 93.95 → 94.0 → A; 93.94 → 93.9 → A–. Any percentage of 94 or higher, including above 100, is an A.

Letter	Percentage	Letter	Percentage	Letter	Percentage
A	94 and above	B–	80 – 82.9	D+	67 – 69.9
A–	90 – 93.9	C+	77 – 79.9	D	63 – 66.9
B+	87 – 89.9	C	73 – 76.9	D–	60 – 62.9
B	83 – 86.9	C–	70 – 72.9	F	below 60

Full policy: `grading_and_class_conduct.md`.

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7. Late Work

No late labs or project deliverables are accepted, under any circumstances. Labs are done and submitted in class.

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8. Attendance

Attendance is expected. Students are responsible for all missed work, notes and materials, regardless of the reason for the absence. The in-class labs are done during the session; the lowest 2 lab scores are dropped.

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9. Accessibility and Accommodations

If you have a documented disability for which accommodations may be required in this class, contact the **Office of Accessible Education (OAE)** — https://www.scu.edu/oae/ — as early as possible to arrange accommodations, and share your accommodation letter with the instructor.

Extra time on timed tests also applies to the in-class labs (30 minutes each): your lab time is extended by the approved amount. Send the instructor your OAE letter **before the first session** so this is set up for every lab.

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10. Academic Integrity and Conduct

- **Your own work.** You may get help from anyone and any resource to understand how to solve your work, but what you submit must be yours alone; copying a solution is plagiarism and will be reported. Exchanging ideas is fine; exchanging detailed solutions is not (`academic_integrity.md`, `grading_and_class_conduct.md`).
- **AI tools.** You may use AI tools (ChatGPT, Claude, Copilot) to learn concepts and debug your own code. What you submit must be your own work; in the project, say where AI-generated code or text was used. Exams are taken in the LockDown Browser, without AI tools.

- **SCU Academic Integrity:** https://www.scu.edu/academic-integrity/
- **Plagiarism:** SCU library guide
- **Discrimination, harassment, and sexual misconduct:** Santa Clara University does not tolerate them. Information, reporting options, and support: https://www.scu.edu/title-ix/.

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11. Getting Help

Need	Where
A question about course content	Ask in class, come to office hours, post on **Camino Discussions** (students, the TA, and the instructor respond), or email the instructor
Installing the course software	Week 0: software setup · `required_software.md`
SQL or Python refreshers	`sql_tutorials.md` · `python_tutorials.md` · `sql_style_guide.md`
LockDown Browser or other SCU technology	SCU Technology Help: (408) 554-5700, or a Computing Support Request

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12. Course Documents

Document	Contents
`course_info_and_desc.md`	Course information, description, learning outcomes
`class_meeting_dates_hours.md`	All quarter dates in one place
`DOWNLOADING_COURSE_FILES.md`	One ZIP per week, no git needed
`grading_and_class_conduct.md`	Grading, letter grades, course policies
`midterm_exam.md` · `final_exam.md`	The exams
`office_hours.md`	Office hours and how to ask questions
`prerequisite.md`	What you need to know before the course
`required_books.md` · `required_software.md`	Books and software
`academic_integrity.md`	Academic integrity
Weekly schedule	Week pages, slides, notebooks, labs
Project guidelines	The team project

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